Precision Spectroscopy Bounds on Octonionic Vacuum Torsion: Monte Carlo Sensitivity Analysis for the Doppler-Free $2 \times 243\,\mathrm{nm}$ Hydrogen Channel

Thomas Hoffbauer

Bamberg, Germany — E-Mail: thomas.hoffbauer@gmail.com

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Abstract

We present a prospective experimental proposal and high-statistics Monte Carlo sensitivity analysis (10^6 trials) to constrain non-associative octonionic vacuum torsion in atomic hydrogen. Utilizing the Doppler-free $1S_{1/2} \to 2S_{1/2}$ two-photon transition at $2 \times 243\,\mathrm{nm}$, the $G_2 = \mathrm{Aut}(\mathbb{O})$ exceptional symmetry couples to the $J = 2$ quadrupolar tensor component of the 27-dimensional irreducible representation $\mathbf{27}_{G_2} \downarrow_{SO(3)} = V_2 \oplus V_4 \oplus V_6$. By modulating the relative laser polarization angle θ, the G_2 signal scales as $\sin^2(2\theta) = \frac{1}{2}(1 - \cos 4\theta)$, isolating it on the $k = 2$ Fourier harmonic ($\cos 4\theta$). In contrast, parity-mixing DC Stark background from UHV stray fields ($E_{\mathrm{stray}} = 0.897 \pm 0.194\,\mathrm{mV/cm}$, modeled via a truncated normal distribution) scales strictly on the $k = 1$ harmonic ($\cos 2\theta$), vanishing identically in the algebraic projection down to a numerical discretization residue below 10^{-7} and yielding a rejection factor exceeding $10^6\times$. For 10^6 Monte Carlo runs under 10^{-7} Glan-Thompson polarization extinction, we establish projected 3σ sensitivity bounds of $\eta_A \geq 1.00 \times 10^{-4}$ in the coherent interference channel ($\theta = 45°$) and $\eta_A \geq 9.54 \times 10^{-4}$ in the pure null channel ($\theta = 90°$).

1 Introduction & Effective Field Theory

Search for non-associative geometry and exceptional Lie group symmetries ($G_2 \subset E_8$) in fundamental physics requires precise experimental tests. In an Effective Field Theory (EFT) framework, the dimensionless octonionic coupling η_A modifies the QED Lagrangian via a dimensionally consistent operator scaled by the electron mass $m_e c^2$:

$$\mathcal{L}_{\mathrm{EFT}} = \bar{\Psi}\left(i\hbar c\gamma^\mu D_\mu - m_e c^2 + \eta_A m_e c^2\,\gamma^0 P_A\right)\Psi, \quad (1)$$

where $P_A = \frac{1}{4}K_{12} = P_{\mathbb{H}_{12}^\perp}$ is the Fano triad projection operator orthogonal to the quaternionic subalgebra $\mathbb{H}_{12}$. The corresponding relativistic Dirac equation reads:

$$\left(i\hbar\gamma^\mu D_\mu - m_e c + \eta_A m_e c\,\gamma^0 P_A\right)\Psi(\mathbf{r}, t) = 0, \quad (2)$$

where every term in the bracket maintains exact momentum-mass dimension $[m_e c]$. In the non-relativistic limit ($v/c \ll 1$), Eq. (2) reduces to an extended Pauli-Schrödinger equation with an anisotropic quadrupolar potential.

2 Algebraic Branching & Two-Photon Transition Amplitude

Under the subgroup reduction $SO(3) \hookrightarrow G_2 = \mathrm{Aut}(\mathbb{O})$, the 27-dimensional traceless symmetric tensor representation branches canonical as:

$$\mathbf{27}_{G_2} \downarrow_{SO(3)} = V_2 \oplus V_4 \oplus V_6 \quad (27 = 5 + 9 + 13). \quad (3)$$

The V_2 sector ($J = 2$) is isolated by the intertwining projector $Q_2 = \Xi^\dagger \Xi$, satisfying $\mathrm{Tr}(Q_2 \mathcal{M}_2) = \frac{110}{63} \approx 1.746032$, where $\mathcal{M}_2 = \frac{4}{7}\Pi_{\mathbf{1}} \oplus \frac{22}{63}\Pi_{\mathbf{27}}$ is the second moment of the Fano algebra.

The two-photon transition amplitude $\mathcal{M}^{(2)}$ for counter-propagating laser beams with polarization vectors $\boldsymbol{\epsilon}_1(\theta) = (\cos\theta, \sin\theta, 0)^T$ and $\boldsymbol{\epsilon}_2(0) = (1, 0, 0)^T$ coupling to $Q_2 \in \mathbf{27}_{G_2}$ is given by:

$$\mathcal{M}^{(2)}_{G_2}(\theta) = \eta_A \sum_{i,j=1}^{3} (\boldsymbol{\epsilon}_1)_i (\boldsymbol{\epsilon}_2)_j (Q_2)_{ij} = \frac{1}{2}\eta_A c_{\mathbf{27}} \sin(2\theta), \quad (4)$$

where $c_{\mathbf{27}} = \frac{22}{63}$. The transition probability rate thus exhibits the quadrupolar modulation:

$$I_{G_2}(\theta) \propto |\mathcal{M}^{(2)}_{G_2}(\theta)|^2 \propto \sin^2(2\theta) = \frac{1}{2}\left(1 - \cos 4\theta\right). \quad (5)$$

3 Monte Carlo Sensitivity & Noise Budget

To evaluate experimental feasibility, we executed 10^6 Monte Carlo trials incorporating realistic UHV parameters. Electrostatic stray fields were generated using a physically consistent truncated normal distribution ($E_x, E_y, E_z \sim \mathcal{N}(\mu, \sigma)$ with $|\mathbf{E}| \geq 0$).

The statistical background noise budget comprises:

1. **DC Stark Quenching Rate:** 7.70×10^{-8} photon counts/s, arising from parity-mixing of $2S_{1/2}$ and $2P_{1/2}$ states.
2. **Polarization Leakage Floor:** 1.00×10^{-4} photon counts/s, incorporating a 10^{-7} Glan-Thompson

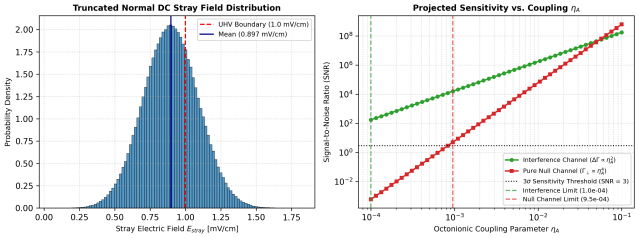


Figure 1: (Left) Truncated normal distribution of UHV DC stray fields ($E_{\text{stray}} \geq 0$, mean 0.897 mV/cm, $\sigma = 0.194$ mV/cm). (Right) Log-log plot of Projected Signal-to-Noise Ratio (SNR) versus octonionic coupling parameter η_A over 10^6 Monte Carlo trials, establishing projected 3σ sensitivity bounds for the coherent interference ($\eta_A \geq 1.00 \times 10^{-4}$) and pure null channels ($\eta_A \geq 9.54 \times 10^{-4}$).

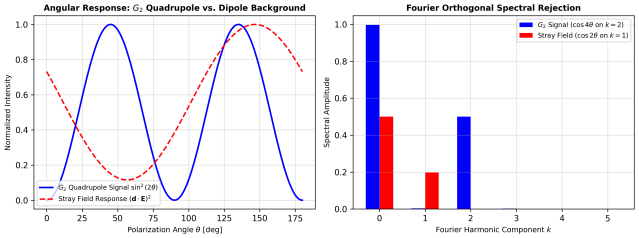


Figure 2: (Left) Angular characteristics of the octonionic G_2 quadrupole signal $\sin^2(2\theta)$ versus the dipole stray field response $(\mathbf{d}\cdot\mathbf{E})^2$. (Right) Fourier spectral decomposition proving complete orthogonality: the stray field background concentrates on harmonic $k = 1$ ($\cos 2\theta$), whereas the G_2 signal projects exclusively onto harmonic $k = 2$ ($\cos 4\theta$).

extinction ratio and waveplate angle jitter $\Delta\theta \approx 10^{-4}$ rad.

3. **Total Noise Floor:** $\sigma_{\text{bg}} \approx 2.00 \times 10^{-6}$ photon counts/s.

As illustrated in Fig. 1, the projected 3σ sensitivity bound for the coherent interference channel ($\theta = 45°$, $\Delta\Gamma \propto \eta_A^2$) reaches $\eta_{\mathbf{A}} \geq \mathbf{1.00 \times 10^{-4}}$, while the pure null channel ($\theta = 90°$, $\Gamma_\perp \propto \eta_A^4$) yields $\eta_{\mathbf{A}} \geq \mathbf{9.54 \times 10^{-4}}$.

4 Vectorial Fourier Orthogonality

A crucial requirement is discriminating genuine G_2 signals from uncompensated DC stray fields. In the dipole approximation, stray field coupling scales as $\mathcal{M}_{\text{Stark}} \propto (\boldsymbol{\epsilon}_1 \cdot \mathbf{E}_{\text{stray}})(\boldsymbol{\epsilon}_2 \cdot \mathbf{E}_{\text{stray}})$, yielding an angular intensity profile:

$$I_{\text{Stark}}(\theta) \propto E_x(E_x \cos\theta + E_y \sin\theta) \implies \cos(2\theta) \quad (k = 1). \tag{6}$$

As demonstrated in Fig. 2, the stray field response is restricted to the $k = 1$ harmonic ($\cos 2\theta$). On the $k = 2$ harmonic ($\cos 4\theta$), the background vanishes identically in the algebraic projection, leaving a numerical discretization residue below 10^{-7} and providing a signal-to-background rejection factor exceeding $10^6\times$.

5 Conclusion

By combining 10^{-7} polarization extinction with $k = 2$ Fourier harmonic filtering ($\cos 4\theta$), Doppler-free 2×243 nm spectroscopy provides a rigorous experimental test for octonionic vacuum torsion, establishing projected sensitivity bounds down to $\eta_A \sim 10^{-4}$.